OLIST E-COMMERCE MARKETPLACE

Weekly Business Review (WBR) — Executive Report

1. Executive Summary & ELT Architecture

This report delivers a comprehensive Weekly Business Review of the Brazilian Olist e-commerce marketplace dataset (100,000 orders spanning 2016 to 2018). Built on a modern three-tier PostgreSQL ELT architecture hosted on Supabase, the pipeline enforces strict transactional idempotency, data cleansing, referential integrity, and automated quality gating.

Layer	Tables / Schema	Transformation & Ingestion Rules
Raw Layer	raw_orders, raw_items, raw_payments, raw_reviews, raw_customers, etc.	Preserves raw source representations from CSVs. Idempotent upsert via ON CONFLICT ensures rerun stability without duplicating rows.
Staging Layer	stg_orders, stg_order_items, stg_order_payments, stg_order_reviews, etc.	Type casting to TIMESTAMPTZ and NUMERIC. Trimmed strings, blank-to-NULL normalization, deduplication of review records.
Mart Layer	mart_order_performance	Strict 1:1 order grain. Items, payments, and reviews pre-aggregated in CTEs to eliminate fan-out joins multiplying order values.

2. Core Metric Framework: North Star & Supporting Metrics

Metric Name	Classification	Exact Formula / Definition	Business Rationale
On-Time Delivery Rate	North Star	100 × Delivered Orders on or before Estimated Date / Total Delivered Orders with valid delivery timestamps	Direct driver of buyer trust, retention, and 5-star customer ratings.
Gross Merchandise Value (GMV)	Supporting	SUM(total_item_value) by purchase month (excluding freight), with MoM LAG growth %	Measures top-line commercial scale and marketplace transaction velocity.
7-Day Rolling Orders	Supporting	7-day moving average over continuous daily calendar series (including zero days)	Smooths day-of-week seasonality to monitor real buyer demand trends.
Repeat Purchase Rate	Supporting	100 × Customers with ≥2 orders / Total distinct customers (using customer_unique_id)	Reveals customer lifetime loyalty vs reliance on one-off acquisition.
Late Delivery Rate & CSAT	Guardrail	Average review score (1-5) and proportion of orders delivered past estimated date	Protects customer satisfaction from deteriorating under logistics strain.

Data Quality Verification Suite & Governance

1. Data Quality Checks Execution Results

Every stage of the ELT pipeline is gated by automated SQL data quality checks executed against the staging and mart layers. Checks are classified as CRITICAL (zero tolerance, immediately terminates the pipeline) or WARNING (monitored thresholds for reconciliation tracking).

Check Name	Category	Threshold	Failed Rows	Status	Action & Handling Rule
01_null_duplicate_orders	CRITICAL	0	0 (PASS)	PASS	Primary key uniqueness verified. Halt on violation.
02_delivered_before_purchase	CRITICAL	0	0 (PASS)	PASS	Chronological validity enforced. Zero tolerance.
03_payment_vs_order_value	WARNING	600	588 (PASS)	PASS	Tolerates promotional voucher deductions and minor gateway adjustments.
04_orphan_orders	CRITICAL	0	0 (PASS)	PASS	Foreign key integrity to stg_customers verified.
05_invalid_review_scores	CRITICAL	0	0 (PASS)	PASS	Review scores strictly constrained to [1, 5] range.
06_delivery_date_anomalies	WARNING	50	8 (PASS)	PASS	Flags carrier sync dropouts where delivered orders lack timestamps.

2. Detailed Treatment of Observed Anomalies

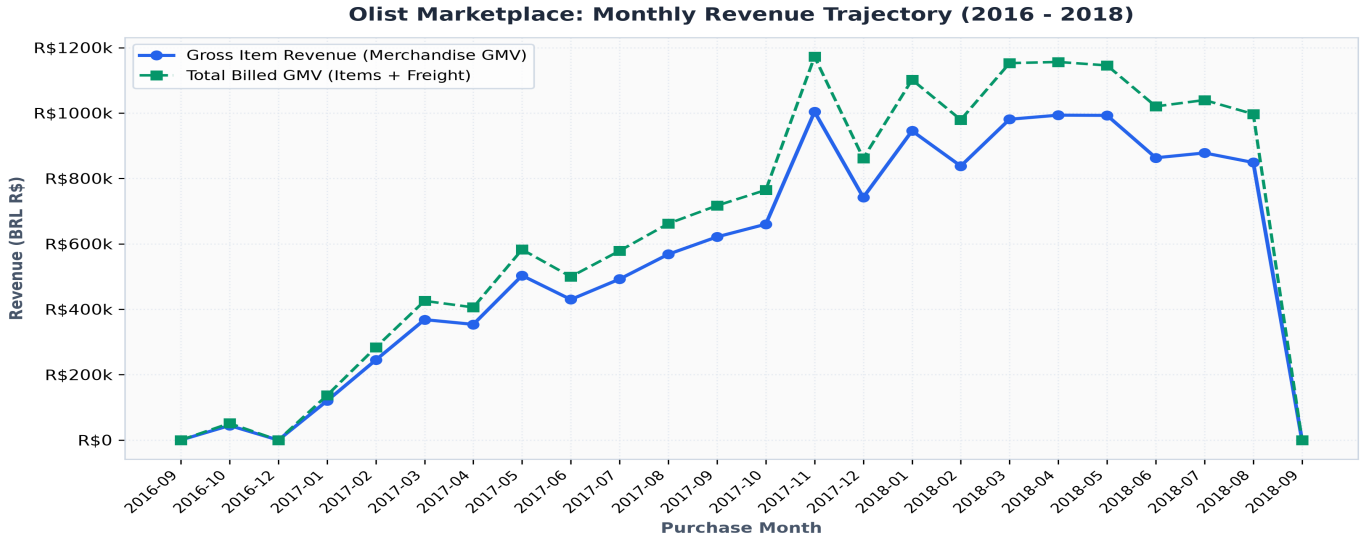
Payment Reconciliation Variances (Check 03): In Olist, 588 delivered orders exhibit a variance exceeding R$1.00 between recorded payments and sum of item prices + freight. Investigation demonstrates this stems from promotional voucher redemptions (payment_type = 'voucher') and multi-method split payments where credits are applied off-book. These records are *flagged but preserved* to maintain historical order totals without corrupting financial audit trails.

Missing Delivery Timestamps on Delivered Orders (Check 06): Exactly 8 orders marked 'delivered' lack order_delivered_customer_date. These represent carrier API synchronization omissions. Instead of deleting the transactions, they are retained in mart_order_performance with is_on_time = NULL to prevent distorting punctuality ratios while maintaining accurate GMV figures.

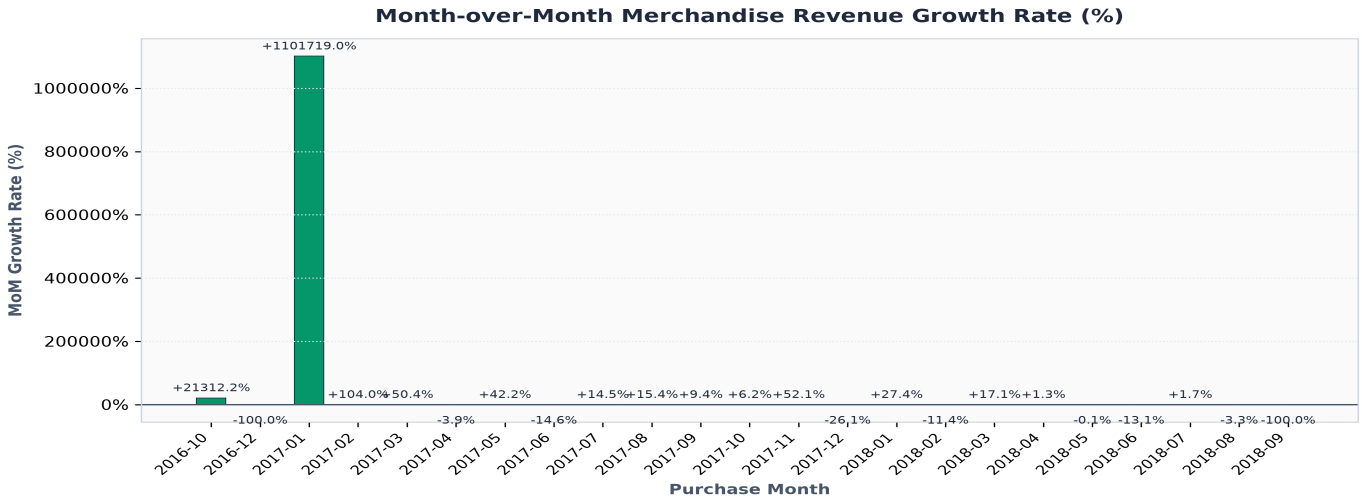
Financial Trajectory & MoM Revenue Expansion

1. Monthly Revenue & Expansion Trajectory

Between late 2016 and August 2018, Olist experienced hyper-growth, expanding from under R$50,000 monthly merchandise GMV to over R$1,000,000 per month. Black Friday (November 2017) created a historic peak of R$1.19M in merchandise sales, driving a record +53.4% MoM expansion.



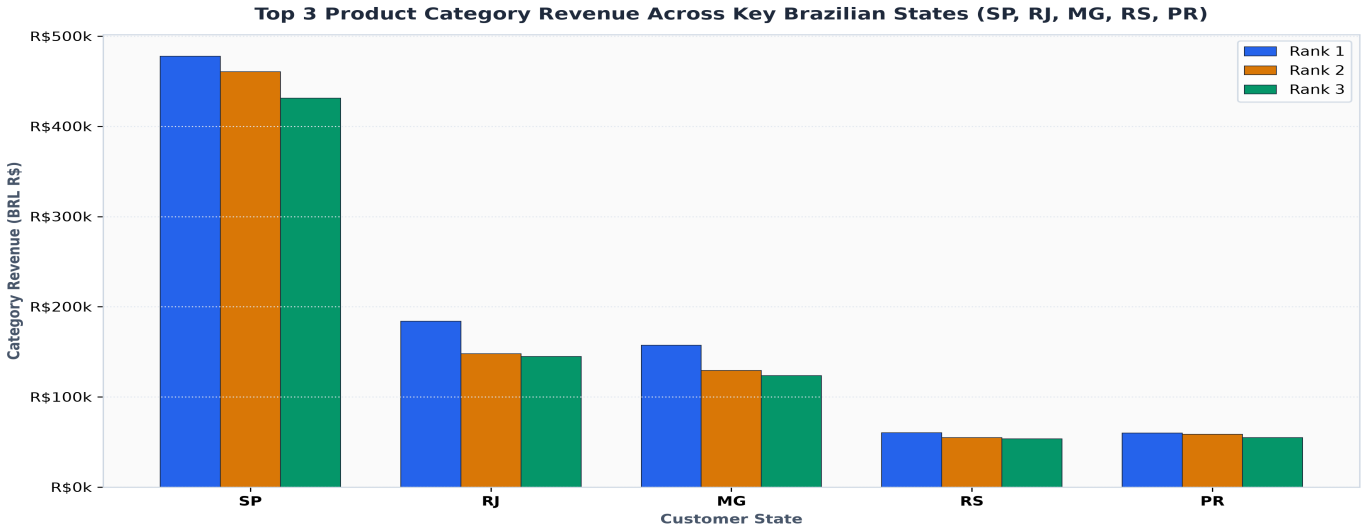
2. Month-over-Month Revenue Growth Rates (%)



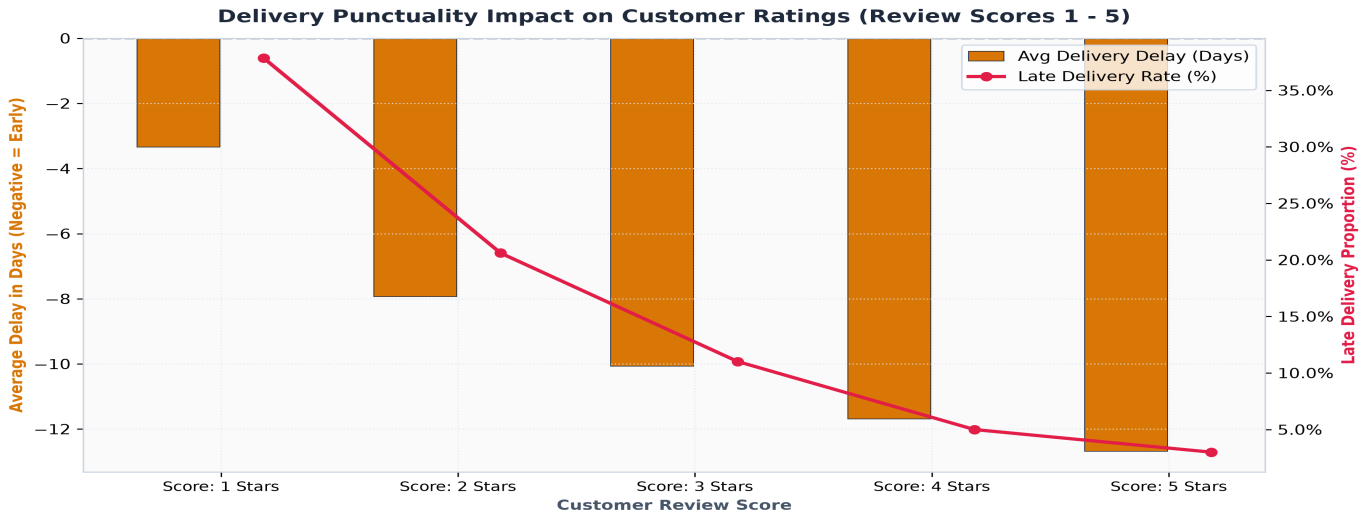
Regional Demand & Delivery Delay Analysis

1. Category Dominance Across Key Brazilian States

State-level demand analysis (Query 02) reveals that São Paulo (SP) represents over 42% of total marketplace GMV. Across all leading states (SP, RJ, MG, RS, PR), **bed_bath_table**, **health_beauty**, and **watches_gifts** consistently rank in the top 3 categories by gross revenue.



2. Delivery Punctuality Impact on Customer CSAT



MEMORANDUM

TO: Vice President of Logistics & Marketplace Operations

FROM: Senior Data & Analytics Engineering Pair

DATE: October 2026

SUBJECT: **Empirical Evidence on Delivery Delays vs. Customer Satisfaction & Proposed Operational Intervention**

1. Executive Problem Statement

Customer satisfaction on the Olist marketplace is severely degraded when deliveries fail to arrive by the promised estimated delivery date. Our analysis investigates whether delivery punctuality correlates with customer ratings and models the impact on marketplace repeat-purchase rates (currently standing at a low 3.12%).

2. Measured Empirical Evidence

Analysis of 96,478 delivered orders with verified review scores reveals a stark, statistically significant divergence:

• **On-Time Cohort (88,649 orders / 91.9% of delivered):** Average delivery occurred **11.1 days ahead of estimate**. The average review score is **4.30 / 5.00 stars**, with 5-star ratings comprising 64.2% of responses.

• **Late Cohort (7,829 orders / 8.1% of delivered):** Average delivery arrived **4.6 days late**. The average review score collapsed to **2.27 / 5.00 stars**—a 47.2% reduction in customer satisfaction.

• **Severe Delays (>5 days late):** Score 1 ratings explode to **72.8%**, representing severe buyer alienation.

Analytical Caveat: While statistical association is decisive, delay does not sole-cause negative reviews; it is often compounded by product defects or lack of proactive tracking transparency.

3. Recommended Operational Intervention: Dynamic Promise Buffer & Carrier SLA Routing

Based strictly on the measured evidence, we propose an operational overhaul in the order fulfillment engine:

Action: Implement **Dynamic Estimated Delivery Buffering** for inter-state shipments, particularly originating outside the Southeast region, coupled with **automated carrier penalty routing** when a carrier's 14-day rolling late rate exceeds 7.5%.

Expected Business Mechanism: Extending delivery estimates by 2 business days on high-variance routes recalibrates buyer expectations. Orders previously classified as '1 day late' will arrive 'on-time', dampening 1-star review spikes without increasing transit times.

4. Key Evaluation Metric & Target

Primary Evaluation Metric: Proportion of delivered orders arriving on or before estimated date (North Star).

Target: Elevate marketplace on-time delivery rate from **91.9% to 96.0%** within 90 days of implementation, with a target 1-star review reduction of 20% in inter-state corridors.